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Faculty of Science
School of Mathematical Sciences

Computational Modelling of Multilateral Treaties

Stage 2

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Supervisory Team

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences, QUT
Associate Supervisor	Prof. Jonathan Rhodes	School of Mathematical Sciences, QUT

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica's Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him deep insight into the policy implications of environmental decisions, providing a grounding for the treaty-system applications explored in this research.

Chapter 1

Background and Literature Review

1.1 Introduction

Consensus regimes in practice reach agreement far more often than their veto structure would suggest. But success is never guaranteed: in bodies such as the United Nations Security Council a single veto routinely blocks collective action, and even historically productive regimes can stall. What separates a regime that keeps deciding from one that stagnates is not well understood. A model that reproduces how consensus is reached or broken is therefore needed, both to explain past behaviour and to anticipate when a regime approaches the limits of its capacity to act.

Models of multi-actor decision-making already exist, but they were not built for this task. Classical opinion-dynamics and game-theoretic models represent each actor with a single, low-dimensional position and are rarely confronted with the actors' own words. Capturing a consensus regime faithfully requires more: a higher-dimensional description in which actors hold distinct positions across many issues, preferences read directly from the text the actors produce and organised by topic, and measures validated and calibrated against observed outputs rather than assumed. Recent advances in natural language processing and machine learning make each of these upgrades feasible.

The Antarctic Treaty System (ATS) is one of those consensus-driven regimes, negotiated among a set of parties called consultative parties (Secretariat of the Antarctic Treaty 2023). This group of countries, together with a second group lacking decision-making power (non-consultative parties), meets yearly at the Antarctic Treaty Consultative Meeting (ATCM). Before each meeting, consultative parties may submit Working Papers (WP) and Information Papers (IP). During ATCMs the parties discuss Working Papers and produce outputs only by consensus. As international relations strain and Antarctic environmental pressures intensify, the risk of losing the capacity to act increases. In

2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges (Gardiner et al. 2025).

1.2 The Antarctic Treaty System as a consensus regime

In the ATS, Consultative Parties (CPs) reach decisions exclusively by consensus. These outputs are decided at the annual ATCM and take three forms (Secretariat of the Antarctic Treaty 2023, Rule 24): **Measures** (adopted texts intended to become legally binding, requiring approval by the relevant parties' legislative bodies), **Decisions** (internal organisational matters), and **Resolutions** (non-binding recommendations or expressions of intent).

There are four types of input:

- **Working Papers** ((Secretariat of the Antarctic Treaty 2023, Rule 48)): submitted by CPs and Observers, these require discussion at the ATCM and are the main source of outputs. They are the only input from CPs that can produce an outcome.
- **Secretariat**, **Information**, and **Background Papers** (Secretariat of the Antarctic Treaty 2023, Rules 49–51): respectively, organisational matters such as the budget; supporting information (also from NCPs and experts); and documents submitted for the record only.

The consensus system makes the ATS a notable example of international cooperation; however, the same mechanism has become a major challenge as the number of CPs grows (Gardiner et al. 2025).

A notable case of contested consensus occurred in the late 1970s and 1980s, during the negotiations over mineral resources in Antarctica. The Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was produced and then abandoned in favour of the Madrid Protocol (1991), which prohibited all mineral resource extraction in Antarctica (Jackson 2021). This well-documented episode provides a valuable canary test: a known case with documented positions against which any quantitative stance measure can be validated.

1.3 Quantitative and text-based study of institutions and treaties

The quantitative study of international institutions long predates its application to Antarctica. Substantial literature maps the design of treaties and organisations into structured variables—for example, their decision rules, dispute-settlement provisions, and enforcement mechanisms. Some studies link these variables to institutional performance, survival, and decline (Gray 2018). A more recent trend treats the *text* of international agreements and debates as data. They recover state positions from voting records and speeches (Baturo et al. 2017; Slapin and Proksch 2008). Both approaches show that preferences and design features of treaty regimes can be measured, not merely described.

The ATS has been analysed quantitatively in two ways: Gardiner et al. (2025) provides performance accounts that measure the regime in aggregate, counting inputs and outputs, and the time legally binding measures take to enter into force. It concludes that responsiveness to challenges in the Antarctic region is declining. The second methodology is structural: it maps who collaborates with whom through co-authorship networks of ATCM documents (Botelho et al. 2026), and how such networks form (Kumar et al. 2026). Both analyses are valuable, and this project uses both as external checks: the network structure as a replication target for the authorship layer, but also the outcome series, by Gardiner et al. (2025), as a prediction target in Chapter 2. Neither, however, extracts topics from WPs in an unsupervised manner.

1.4 Measuring policy preference from political text

Positional measures extracted from political texts are a well-developed field with two paradigms. The first paradigm is scaling and ideal-point methods, which recover a latent positional dimension without a pre-specified axis—there is no need to fix, say, a regulatory-versus-permissive comparison in advance. These methods typically rely on word counts (i.e. frequencies) as their main underlying principle. Wordfish (Slapin and Proksch 2008) is an unsupervised positional scaling of party text. Lauderdale and Herzog (2016) addressed the problem that legislators talk about different things in different debates; they introduced a per-topic, per-debate scaling. Vafa et al. (2020) produced a probabilistic framework that jointly places ideal points and models actors' language. The closest dataset to the ATCM corpus is the one Baturo et al. (2017) use to derive state-level preferences from United Nations (UN) debate speeches. Finally, a recent review by Parschan and Jakob (2025) condenses all these approaches in one paper.

The second paradigm uses embeddings as its main tool. Rheault and Cochrane (2019) show that word-embedding geometry recovers ideological placement using parliamentary data. Rodriguez and Spirling (2022) provide a comprehensive analysis of what works and what does not when analysing political text using embedding techniques. They find embeddings a useful technique for those purposes but caution that the choice of embeddings strongly affects applied inference, and that the results must be validated.

This thesis uses the second paradigm in the same pipeline at different stages. It also differs from the ideal-point methods in an important way: ideal-point models recover a single stable latent position per actor, whereas this project requires a topic-conditioned stance, allowing parties to hold different positions on different topics through time. The topic structure is discovered using BERTopic (Grootendorst 2022) and the stance with discriminative transformer-NLI models (Burnham 2023; Abercrombie and Batista-Navarro 2022). The results can be stress-tested directly using the CRAMRA canary test (Jackson 2021), or indirectly by checking which countries cluster around $(\text{topic}, \text{stance})$ pairs and comparing this to the Botelho et al. (2026) network results.

1.5 Modelling consensus: responsive and anticipating approaches

There are two traditional groups of models for how groups reach consensus. The first group is responsive: each actor reacts to the current positions of others. These are the synchronization and opinion-dynamics models; examples are the Kuramoto and opinion-changing-rate models used by Botelho et al. (2026), which represent agreement as a coupled-oscillator convergence. These methods treat consensus as *convergence*. DeGroot (1974) iteratively average agents' opinions using $x(t+1) = Wx(t)$ until they converge. A bounded-confidence version was later developed by Hegselmann and Krause (2002). A further model in this group is the stubborn-agents model (Friedkin and Johnsen 1990), in which each agent only partially updates toward its neighbours and stays anchored to its initial opinion. This group contains no strategic preference, no veto, and no learning; it therefore cannot simulate the bargaining behaviour of consensus-based systems.

The second group is anticipating: each actor reasons about how the others will respond. These models use social choice and game theory to treat consensus as a unanimity rule. Tsebelis (2002) develop a veto-players framework in which every party has veto power, so the status quo is replaced only when $u_i(p) \geq u_i(q)$ for every party i. Consequently, as the number of parties increases and their positions spread, the set of adoptable policies shrinks and the system becomes harder to change. Gardiner et al. (2025) reach the same conclusion when analysing the ATS.

Between these two groups lies a middle ground: models that retain the strategic, veto-bearing structure of the second group but, instead of solving for equilibrium in closed form, learn it from repeated interaction. This middle ground has grown into a field of its own and is treated separately in the next section.

1.6 Negotiation and coalition formation simulation using reinforcement learning

The strategic view of consensus developed above implies that, in consensus-based systems, the best way to replicate behaviour among many heterogeneous Parties, each holding a veto and bargaining over a status quo, is to use repeated strategic interaction. In the specific case of the quantitative simulation of the ATS over the years, its size and complexity mean that the closed-form equilibria of analytic game theory are intractable. In these cases, multi-agent reinforcement learning (MARL) is the appropriate tool, precisely because it keeps the game-theoretic substance while relaxing this constraint. Formally, MARL is defined over stochastic (Markov) games, a generalisation of repeated and normal-form games. Similarly, its learning procedures approximate equilibrium behaviour from interaction rather than solving for it in closed form (Littman 1994). In this context, neural networks (NN) enter only as function approximators that allow this learning to scale to large state and action spaces.

Foundational MARL work covers multi-issue bargaining (Chang 2021), negotiating-team formation (Bachrach et al. 2020), bargaining under risk and asymmetric information (Schmid et al. 2020), and search-improved game-theoretic learning (Li et al. 2023). Two reviews cover the field: Gronauer and Diepold (2022) on MARL and Sarkar et al. (2022) on coalition formation. Most recently, MARL agents based on Large Language Models (LLMs) have been used to model political negotiation directly: for example, Moghimifar et al. (2024) simulate coalition formation in parliament using text data from party manifestos. Bolleddu (2025) present a MARL system for consensus building and negotiation based on synthetic data. Kulkarni et al. (2025) use negotiated agreements with game-theoretic reasoning to predict dynamic coalition formation in Diplomacy (a well-known benchmark for this type of simulation). Finally, some other benchmarks for real negotiation data are starting to appear (Benac et al. 2026). However, these simulations run on synthetic data or abstract benchmarks rather than the deliberative records of a real consensus regime.

1.7 Summary of literature gap

The descriptive work shows what the regime outputs but not who produces what. Structural and party-network analyses are undirected and unattributed. Text-based analysis recovers latent positions rather than topic-specific stance over time. Finally, consensus has been simulated either as convergence or, where learning-based methods were used, on synthetic data rather than the institution's own deliberative records. These literatures differ in method but none recovers a directional preference: who is for or against what extracted from the regime's own words.

This gap in the existing research has a cost because no such measure exists. Therefore, the decline in responsiveness and the 2022 consensus failure introduced in Section 1.1 cannot be attributed. This means the field cannot yet tell whether they are driven by particular parties or are structural to the consensus rule itself. Recovering that measure is what would let the field answer this question.

No prior work extracts a directional preference for each party over time from a consensus regime's own textual records, nor uses such preferences to drive a strategic simulation in which agents have veto power. This thesis supplies both halves and joins them: the preferences recovered from the record become the parameters of the simulation that reproduces consensus and its failures.

Chapter 2

Program and design of the research project

2.1 Research questions

The thesis addresses the gap mentioned above by answering these research questions:

Research Questions

RQ1. Can parties' strategic positions be reconstructed from written records into a directional measure across topics and time?

RQ2. Do the reconstructed positions recover known history and predict outcomes?

RQ3. Can a multi-agent model parameterised by it reproduce consensus and test counterfactual decision rules?

2.2 Objectives, methodology and research plan

Figure 2.1 shows the overall flow of the research project, the output of each stage, and how the objectives relate to one another. Throughout the following sections each chapter is described in detail.

2.2.1 Objective 1 / Chapter 1: The Support layer

The **aim** is to produce a defensible, directional measure of parties' preferences over relevant topics through time, using documents the parties submit themselves. For

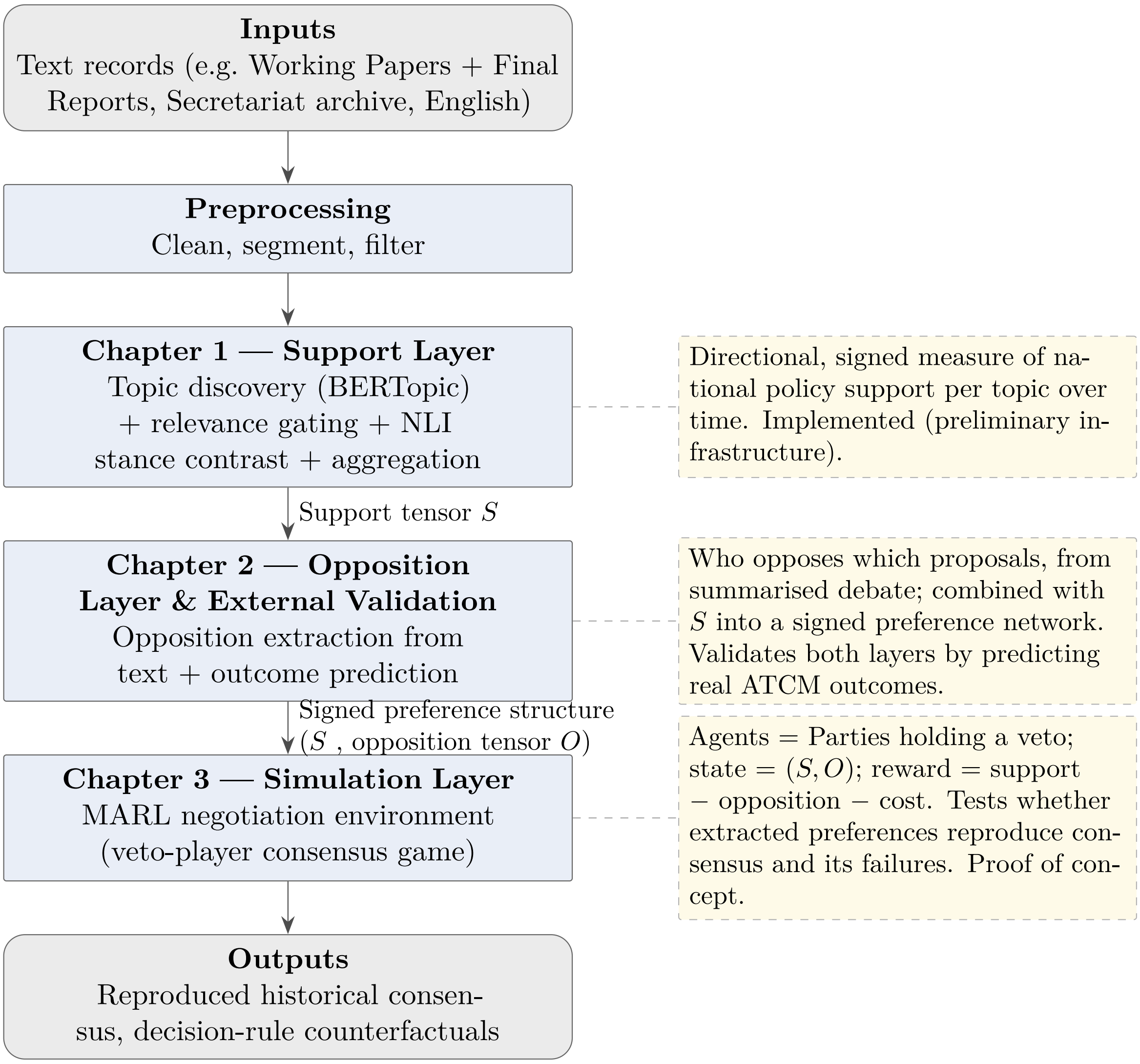

Figure 2.1: Thesis workflow: from text records to a validated consensus simulation.

the ATCM case, Working Papers (WPs) are the input data because, of all submitted documents, these are the only ones submitted by parties or Observers (non-Secretariat) that require discussion and action. This makes WPs the most likely text to contain parties' strategic intentions. The analysis uses English-language papers, since these represent the largest share of all WPs. It is also relevant that most of the available pre-trained embedding and NLI models were trained on English-language data.

The **approach** is a pipeline with multiple stages. The first set of stages applies BERTopic (Grootendorst 2022) to the whole corpus, producing a set of topics K with corresponding cluster centroids $\{\boldsymbol{\mu}_k\}_{k\in K}$, where each $\boldsymbol{\mu}_k \in \mathbb{R}^m$ in the m-dimensional embedding space. The second set of stages selects, for each document and topic, the passages relevant to that topic. Each document d is split into passages $P_d = \{p_1, \ldots, p_{n_d}\}$, and each passage p is embedded as $\mathbf{e}_p \in \mathbb{R}^m$. Relevance acts as a gating mechanism that filters out irrelevant passages: for each document d and topic k, every passage $p \in P_d$ is scored by its cosine similarity to the topic centroid as follows,

$$\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k) = \frac{\mathbf{e}_p \cdot \boldsymbol{\mu}_k}{\|\mathbf{e}_p\| \, \|\boldsymbol{\mu}_k\|}.$$

$P_{d,k}$ is the set of top scoring passages above a threshold. Then, the relevant sentences out of each $P_{d,k}$ are extracted using a Large Language Model (LLM). These sentences are concatenated and scored as a set by the classifier. It produces one three-class distribution which is mapped to five ordinal classes that range from strong opposition to strong support. Formally, it is as follows

$$\pi_{d,k} = \mathrm{NLI}\Big(\mathrm{Extract}(P_{d,k})\Big) \in \Delta_5, \qquad \Delta_5 = \Big\{ \pi \in \mathbb{R}^5 \;\Big|\; \pi_\ell \geq 0,\ \textstyle\sum_{\ell=1}^5 \pi_\ell = 1 \Big\}, \tag{2.1}$$

The signed stance of document d on topic k is the expected value of this distribution under an equal-spacing convention. This is:

$$s_{d,k} = \sum_{\ell=1}^{5} v_\ell \, \pi_{d,k}(\ell), \qquad v = \Big(-1,\ -\tfrac{1}{2},\ 0,\ \tfrac{1}{2},\ 1 \Big). \tag{2.2}$$

Since $\pi_{d,k}$ is a probability distribution and every $v_\ell \in [-1, 1]$, the stance $s_{d,k} \in [-1, 1]$ is a convex combination of the class values and therefore well defined on topic k's axis.

Finally, the previous result is aggregated to produce *country* $\times$ *topic* $\times$ *year* stances over the Working Papers weighted by co-sponsorship credit:

$$S_{c,k,y} = \sum_{\substack{d\in D(c,y) \\ k\in K(d)}} w_{d,c} \, s_{d,k}, \tag{2.3}$$

where $D(c, y)$ is the set of Working Papers co-authored by country c in year y, $K(d)$ is the set of topics assigned to document d, and $w_{d,c}$ is the attribution weight of that country in one document. Two attribution conventions are computed:

- full credit, $w_{d,c} = 1$, which means every co-sponsor receives the full position and which answers "who is on this side?".
- fractional credit, $w_{d,c} = 1/|A_d|$ with A_d the set of co-sponsors of d, which answers "who is driving this?".

For a fixed year, the slice $S_{\cdot,\cdot,y} \in \mathbb{R}^{C\times|K|}$ is a signed *country* $\times$ *topic* matrix, with C countries and $|K|$ topics. See Appendix C for full notation.

Measuring directional stance on diplomatic text using the distribution in (2.1) is challenging, because hedging and similar prose obscure positioning. In other words, permissive and restrictive positions on a regulatory question share the same safeguard vocabulary; scoring a single hypothesis therefore collapses toward neutrality. Therefore, stance is recovered as a *contrast* between entail and contradiction positions. Specifically, for each topic k, structured hypotheses anchor a for-pole h_k^+, a neutral one h_k^0, and an against one h_k^- on the topic's contestable axis. These $(h_k, premise)$ pairs are input into NLI classifier (Laurer et al. 2024). The resulting scores for *entailment* and *contradiction* are subtracted. The classifier's results are cross-checked against a generative model used as a secondary instrument (Le Mens and Gallego 2025).

Validation strategy The construct will be validated using three independent checks, each targeting a different stage of the pipeline. A topic validation check compares the discovered topics against the Secretariat's categories and the WP in each category. A historical canary uses the minerals/CRAMRA negotiations of the late 1970s and 1980s as documented ground truth. The test checks whether the pipeline recovers the known directional opposition in those papers. Finally, a network comparison check reconstructs the co-authorship structure reported by Botelho et al. (2026) from the authorship layer.

Figure 2.2 below gives a summarised overview of the steps described above.

2.2.2 Objective 2 / Chapter 2: The Opposition layer and external validation

The **aim** of this part of the research project is to extract a directional opposition signal. This opposition signal has two sides. The first is opposition to the issue itself; it is not the inverse of the support-layer result, but rather the *veto signal* against a given proposal. The second side is a secondary, exploratory result: a measure of a country's

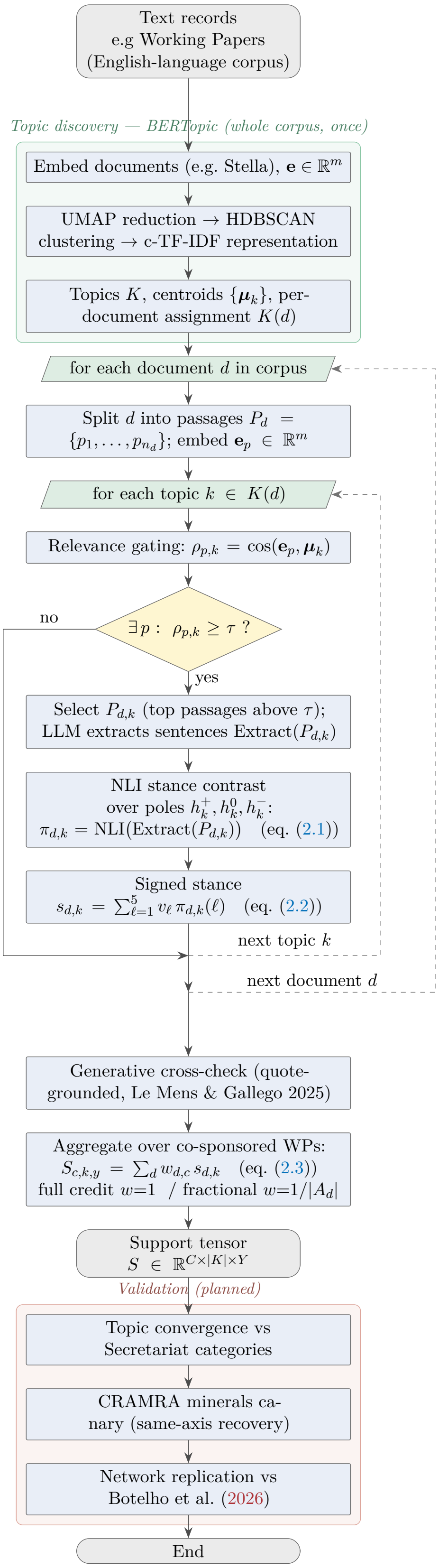

Figure 2.2: Chapter 1 support-layer pipeline, from text records to the country×topic×year support tensor S.

general propensity to reject proposals within a given ATCM. As with the support layer, this chapter builds a pipeline and, for the specific case of the ATS, the ATCMs' final reports are used as text input. Given the scarcity of text in which countries veto a proposal, the attribution is expected to be harder than for the support layer. Each extracted statement yields a *(country, target, polarity)* triple, with rationale and source quote. Taking the topics K from Chapter 1 and mapping targets back to them yields an opposition tensor $\mathbf{O}$ with the same Country $\times$ Topic $\times$ Year structure. At this point, there is a pair $(\mathbf{S}, \mathbf{O})$ of tensors that constitutes a signed preference network. These two tensors are the inputs to the third chapter.

External validation. The combined structure (the tensor pair $(\mathbf{S}, \mathbf{O})$) is tested for predictive content against observed outcomes (see Section 1.2) and the compilation by Gardiner et al. (2025), for the case of the ATS. This gives Chapter 2 an independent research question, tests the results so far, and produces a quantitative description. The novelty in this work is the directional signed signal derived from the institution's own input records and outcomes. Although this thesis focuses on the ATS, the pipeline is institution-agnostic: the same $(\mathbf{S}, \mathbf{O})$ construction could be applied to other consensus- or veto-based international institutions whose deliberative records are available.

The main risks in this chapter are the sparsity of text that can be directly attributed to an input, and the identification of the country that vetoes any proposal. A further limitation is the influence of external geopolitics, as in the case of the consensus breakdown in 2022 due to the war in Ukraine (Gardiner et al. 2025). The cause of this type of exogenous influence is unlikely to be reflected in the text data of an institution not directly related to such issues.

Figure 2.3 sketches the opposition layer pipeline and its external-validation step for the ATCM case study.

2.2.3 Objective 3 / Chapter 3: The Simulation layer

The **aim** of this chapter is to simulate consensus processes with country-agents whose state is parameterised by the empirically derived support and opposition from previous chapters. Its main measure of success is to be able to reproduce historical trajectories, and examine institutional counterfactuals. In consensus-based systems every Party is a veto player (Tsebelis 2002) and the representation is a strategic bargaining game over a status quo rather than an averaging dynamic.

The **approach** to tackle this task is to specify a custom negotiation environment in which proposals are presented one at a time. Each proposal is an input text/Working

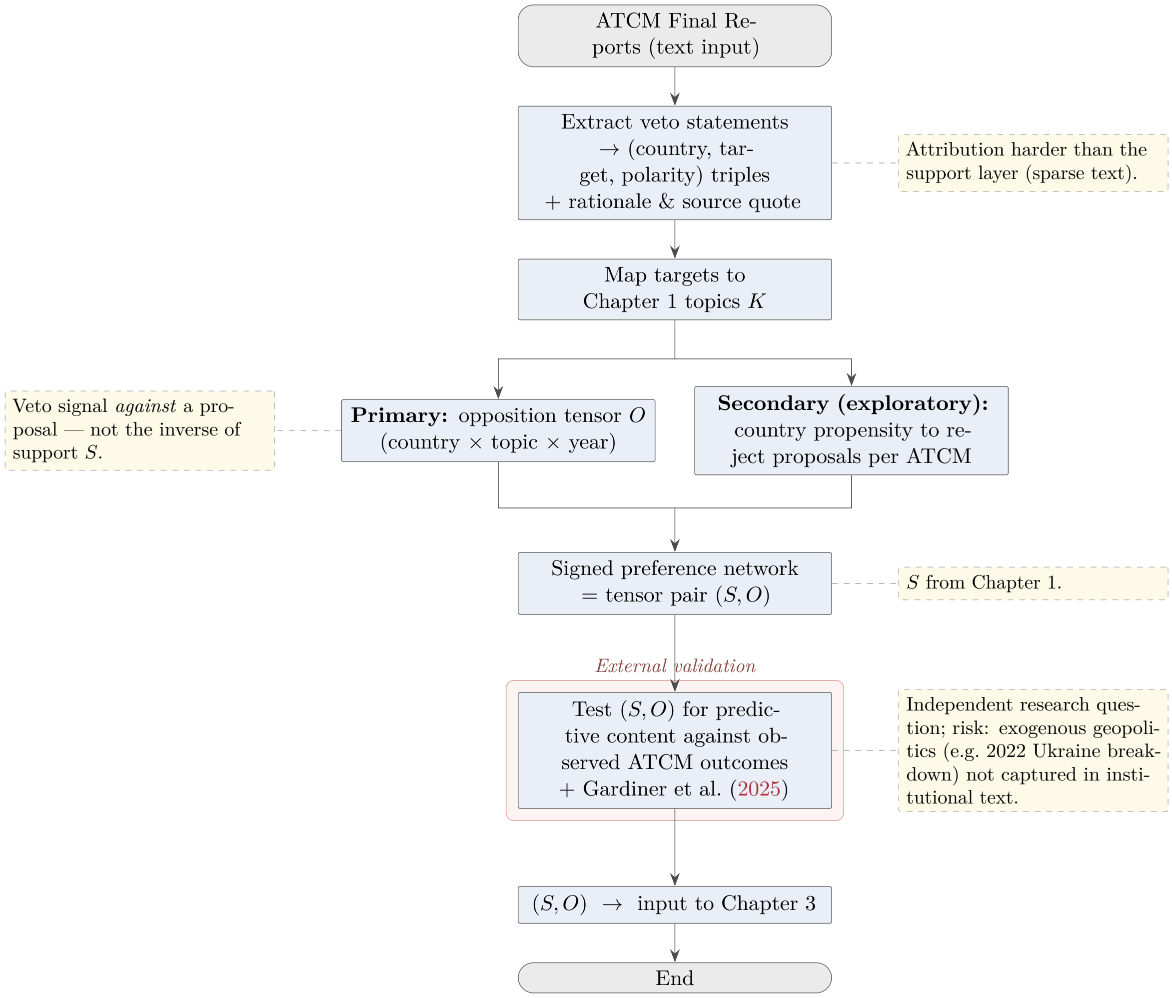

Figure 2.3: Chapter 2 opposition layer sketch: from text data to opposition tensor O, forming the signed preference network (S, O).

Paper d that is mostly about topic k in year y. Then the agents (one per consultative party) decide whether or not to object to a proposal. Each agent's decision-making is supported by its support vector (Chapter 1), its opposition history (Chapter 2), and metadata such as consultative status, official language, and GDP, used as context features rather than as an independent contribution. An agent's action on a presented proposal is binary: to object or not, $o_i \in \{0, 1\}$. Under the consensus rule the proposal is adopted only if no Party objects,

$$1_{\text{adopt}} = \prod_j \left(1 - o_j\right), \tag{2.4}$$

therefore, a single objection vetoes adoption. The reward to agent i is then

$$R_i = \alpha\, S_{i,k,y}\, 1_{\text{adopt}} - \beta\, O_{i,k,y}\, 1_{\text{adopt}} - \gamma\, o_i, \tag{2.5}$$

where the first term rewards adoption of outcomes the Party supports. The second penalises adoption of outcomes it opposes. The third subtracts a diplomatic cost for casting an objection. Finally, $\alpha, \beta, \gamma \geq 0$ are non-negative trade-off weights. R_i is measured relative to the status quo and normalised to zero. For a non-zero 1_{adopt} all parties need to agree (non-object), therefore, every agent holds a genuine veto. Furthermore, consensus emerges from the strategic trade-off between blocking a proposal and bearing the cost of objecting.

The model is validated by the predicted consensus or failure and by an ablation that removes different layers to quantify its contribution. Then a counterfactual analysis (RQ3) examines alternative decision rules, for example, qualified majority in place of unanimity, which relaxes (2.4). This counterfactual intends to predict the behaviour of the system given a change in the institutional rules e.g. ATCM framework (Secretariat of the Antarctic Treaty 2023). This is compared against the effectiveness metrics of Gardiner et al. (2025). This chapter trains its agents with empirically extracted positions from the institution's own record. This methodology makes this the highest-risk chapter and is framed as a proof of concept, contingent on Chapters 1 and 2 passing validation.

2.3 Resources and funding required

The project uses openly published data from the Antarctic Treaty Secretariat document and may use other public data by other international institutions. The project uses open-source software and/or QUT-licensed software. Computation is served by the QUT high-performance computing cluster. No additional funding is required.

2.4 Individual contribution to the research team

I am the sole researcher on this project, with guidance from my supervisory team. The Chapter 1 infrastructure developed at the outset of candidature is preliminary work that establishes feasibility. This is not presented in this document as thesis results, and all reported findings will be produced within candidature. The thesis extends prior group work by Botelho et al. (2026), Kumar et al. (2026) and Carter et al. (2026).

2.5 Timeline for completion

The candidature is full-time and commenced on the 31st of March, 2026. All milestones are placed against the QUT full-time schedule (Table 2.1).

Table 2.1: Candidature timeline (April 2026–April 2029). Ch. 1 Ch. 2 Ch. 3 Milestones/events.

Activity	**3** Jul '26	**6** Oct '26	**9** Jan '27	**12** Apr '27	**15** Jul '27	**18** Oct '27	**21** Jan '28	**24** Apr '28	**27** Jul '28	**30** Oct '28	**33** Jan '29	**36** Apr '29
Milestones												
Stage 2												
Confirmation												
Progress Report												
Final Seminar												
Thesis Writing & Submission												
Objective 1 / Chapter 1: The Support Layer												
Pipeline Development												
Analysis & Validation												
Writeup & Submission												
Objective 2 / Chapter 2: The Opposition Layer												
Pipeline Development												
Extraction & Validation												
Writeup & Submission												
Objective 3 / Chapter 3: The Simulation Layer												
Environment Design												
Training & Evaluation												
Writeup & Submission												

Table 2.1 (continued) — Conference Attendance and Coursework.

Activity	**3** Jul ’26	**6** Oct ’26	**9** Jan ’27	**12** Apr ’27	**15** Jul ’27	**18** Oct ’27	**21** Jan ’28	**24** Apr ’28	**27** Jul ’28	**30** Oct ’28	**33** Jan ’29	**36** Apr ’29
Conference Attendance												
QANZIAM 2026												
AMEG Symposium 2026												
QANZIAM 2027												
International Conference												
AMEG Symposium 2027												
QANZIAM 2028												
Coursework												
RIO (completed)												
AIRS / IFN006												

Coursework (IFN006 Advanced Information Research Skills) is currently underway and will be completed by July 2026 following Stage 2 submission. The Research Integrity Online quiz is completed (see Appendix A).

References

Abercrombie, Gavin and Riza Batista-Navarro (2022). “Policy-focused stance detection in parliamentary debate speeches”. In: *Northern European Journal of Language Technology* 8.1. DOI: 10.3384/nejlt.2000-1533.2022.3264.

Bachrach, Yoram, Richard Everett, Edward Hughes, Angeliki Lazaridou, Marc Lanctot, Guy Lever, Michael Vlassopoulos, and Thore Graepel (2020). “Negotiating team formation using deep reinforcement learning”. In: *Artificial Intelligence* 288, p. 103356. DOI: 10.1016/j.artint.2020.103356.

Baturo, Alexander, Niheer Dasandi, and Slava J. Mikhaylov (2017). “Understanding state preferences with text as data: Introducing the UN General Debate corpus”. In: *Research & Politics* 4.2. DOI: 10.1177/2053168017712821.

Benac, Leo, Jonas Raedler, Zilin Ma, and Finale Doshi-Velez (2026). *A Benchmark for Multi-Party Negotiation Games from Real Negotiation Data*. DOI: 10.48550/arXiv.2603.14066. arXiv: 2603.14066.

Bolleddu, Divya (2025). *Dialogue Diplomats: An end-to-end multi-agent reinforcement learning system for automated conflict resolution and consensus building*. DOI: 10.48550/arXiv.2511.17654. arXiv: 2511.17654.

Botelho, Larissa Lubiana, Kate J. Helmstedt, Kerrie A. Wilson, and Michael Bode (2026). “Coalition Dynamics in Antarctic Governance”. Unpublished manuscript.

Burnham, Michael (2023). “Stance detection: A practical guide to classifying political beliefs in text”. In: *Political Science Research and Methods*. DOI: 10.1017/psrm.2024.35.

Carter, Zachary T., Larissa Lubiana Botelho, Steven L. Chown, Parsa Yaryab, Nadiah P. Kristensen, Kerrie A. Wilson, Daniela Liggett, and Michael Bode (2026). “The Limits of Antarctic Conservation Governance”. Manuscript submitted for publication.

Chang, Ho-Chun Herbert (2021). “Multi-issue negotiation with deep reinforcement learning”. In: *Knowledge-Based Systems* 211, p. 106544. DOI: 10.1016/j.knosys.2020.106544.

DeGroot, Morris H. (1974). “Reaching a consensus”. In: *Journal of the American Statistical Association* 69.345, pp. 118–121. DOI: 10.1080/01621459.1974.10480137.

Friedkin, Noah E. and Eugene C. Johnsen (1990). "Social Influence and Opinions". In: *Journal of Mathematical Sociology* 15.3–4, pp. 193–206. DOI: 10.1080/0022250X.1990.9990069.

Gardiner, Nicholas B., Natasha Gilbert, Daniela Liggett, and Michael Bode (2025). "Measuring the performance of Antarctic Treaty decision-making". In: *Conservation Biology* 39, e14349. DOI: 10.1111/cobi.14349.

Gray, Julia (2018). "Life, Death, or Zombie? The Vitality of International Organizations". In: *International Studies Quarterly* 62.1, pp. 1–13. DOI: 10.1093/isq/sqx086.

Gronauer, Sven and Klaus Diepold (2022). "Multi-agent deep reinforcement learning: A survey". In: *Artificial Intelligence Review* 55, pp. 895–943. DOI: 10.1007/s10462-021-09996-w.

Grootendorst, Maarten (2022). *BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure*. arXiv: 2203.05794 [cs.CL]. URL: https://arxiv.org/abs/2203.05794.

Hegselmann, Rainer and Ulrich Krause (2002). "Opinion Dynamics and Bounded Confidence: Models, Analysis and Simulation". In: *Journal of Artificial Societies and Social Simulation* 5.3. URL: https://www.jasss.org/5/3/2.html.

Jackson, Andrew (2021). *Who Saved Antarctica?* Cham: Palgrave Macmillan. ISBN: 978-3-030-78404-1. DOI: 10.1007/978-3-030-78405-8.

Kulkarni, Abhishek N., Andy Liu, Jean-Raphaël Gaglione, Daniel Fried, and Ufuk Topcu (2025). "Dynamic Coalition Structure Detection in Natural Language-based Interactions". In: *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. Detroit, MI, USA: IFAAMAS, pp. 1227–1234.

Kumar, Viney, Valeria Senigaglia, Maria Kleshnina, and Michael Bode (2026). "Consistent Patterns of Interaction Within Antarctic Governance Forums". Unpublished manuscript.

Lauderdale, Benjamin E. and Alexander Herzog (2016). "Measuring political positions from legislative speech". In: *Political Analysis* 24.3, pp. 374–394. DOI: 10.1093/pan/mpw017.

Laurer, Moritz, Wouter van Atteveldt, Andreu Casas, and Kasper Welbers (2024). *Building Efficient Universal Classifiers with Natural Language Inference*. DOI: 10.48550/arXiv.2312.17543. arXiv: 2312.17543 [cs.CL]. URL: https://arxiv.org/abs/2312.17543.

Le Mens, Gaël and Aina Gallego (2025). "Positioning Political Texts with Large Language Models by Asking and Averaging". In: *Political Analysis* 33.3, pp. 274–282. DOI: 10.1017/pan.2024.29.

Li, Zun, Marc Lanctot, Kevin R. McKee, Luke Marris, et al. (2023). *Search-improved game-theoretic multiagent reinforcement learning in general and negotiation games*. arXiv preprint. arXiv: 2305.11607.

Littman, Michael L. (1994). "Markov Games as a Framework for Multi-agent Reinforcement Learning". In: *Proceedings of the Eleventh International Conference on Machine Learning (ICML)*, pp. 157–163.

Moghimifar, Farhad, Yuan-Fang Li, Robert Thomson, and Gholamreza Haffari (2024). *Modelling political coalition negotiations using LLM-based agents*. DOI: 10.48550/arXiv.2402.11712. arXiv: 2402.11712.

Parschan, Patrick and Charlott Frederike Jakob (2025). "Computational measurement of political positions: A review of text-based ideal point estimation algorithms". In: *Quality & Quantity*. DOI: 10.1007/s11135-025-02500-4. arXiv: 2511.13238.

Rheault, Ludovic and Christopher Cochrane (2019). "Word embeddings for the analysis of ideological placement in parliamentary corpora". In: *Political Analysis* 28.1, pp. 112–133. DOI: 10.1017/pan.2019.26.

Rodriguez, Pedro L. and Arthur Spirling (2022). "Word embeddings: What works, what doesn't, and how to tell the difference for applied research". In: *The Journal of Politics* 84.1, pp. 101–115. DOI: 10.1086/715162.

Sarkar, Samriddhi, Mariana Curado Malta, and Animesh Dutta (2022). "A survey on applications of coalition formation in multi-agent systems". In: *Concurrency and Computation: Practice and Experience* 34.11, e6876. DOI: https://doi.org/10.1002/cpe.6876.

Schmid, Kyrill, Lenz Belzner, Thomy Phan, Thomas Gabor, and Claudia Linnhoff-Popien (2020). "Multi-agent Reinforcement Learning for Bargaining under Risk and Asymmetric Information". In: pp. 144–151. DOI: 10.5220/0008913901440151.

Secretariat of the Antarctic Treaty (2023). *Rules of Procedure of the Antarctic Treaty Consultative Meeting and the Committee for Environmental Protection*. Updated August 2023. Buenos Aires: Secretariat of the Antarctic Treaty. 46 pp. ISBN: 978-987-8929-23-1. URL: https://www.ats.aq (visited on 06/09/2026).

Slapin, Jonathan B. and Sven-Oliver Proksch (2008). "A scaling model for estimating time-series party positions from texts". In: *American Journal of Political Science* 52.3, pp. 705–722. DOI: 10.1111/j.1540-5907.2008.00338.x.

Tsebelis, George (2002). *Veto Players: How Political Institutions Work*. Princeton, NJ: Princeton University Press.

Vafa, Keyon, Suresh Naidu, and David M. Blei (2020). "Text-based ideal points". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 5345–5357. DOI: 10.18653/v1/2020.acl-main.475.

Appendix A

Appendices

Appendix A: Research Integrity Online

[Insert RIO completion certificate. I am on it, next I will add it to this document before submission]

Appendix B: Coursework

AIRS / IFN006 is underway, this course will end on july.

Appendix C: Notation

Indices and sets.	c, i	countries (Consultative Parties), $c \in \{1, \ldots, C\}$
	k	topic, $k \in K$; $\lvert K \rvert$ is the number of topics
	y	year
	d	document (Working Paper)
	p	passage (chunk) of a document
	ℓ	ordinal stance class, $\ell \in \{1, \ldots, 5\}$
	$D(c, y)$	Working Papers co-sponsored by country c in year y
	$K(d)$	topics assigned to document d
	P_d	passages of document d, $P_d = \{p_1, \ldots, p_{n_d}\}$
	$P_{d,k}$	passages of d selected as relevant to topic k
	A_d	co-sponsors of document d

Embedding space.

m	embedding dimensionality (full Stella space)
$\mathbf{e}_p$	embedding of passage p, $\mathbf{e}_p \in \mathbb{R}^m$
$\boldsymbol{\mu}_k$	centroid of topic k, $\boldsymbol{\mu}_k \in \mathbb{R}^m$
$\rho_{p,k}$	passage–topic relevance, $\rho_{p,k} = \cos(\mathbf{e}_p, \boldsymbol{\mu}_k)$
τ	relevance selection threshold

Stance and hypotheses.

h_k^+, h_k^0, h_k^-	for-, neutral-, against-pole hypotheses for topic k
$\lambda(d, h)$	null-premise-debiased entailment log-ratio (DCPMI)
$\tilde{\pi}_{d,k}$	three-pole distribution (stance classifier)
$\pi_{d,k}$	five-class ordinal distribution, $\pi_{d,k} \in \Delta_5$
Δ_5	probability simplex, $\{\pi \in \mathbb{R}^5 : \pi_\ell \geq 0,\ \sum_\ell \pi_\ell = 1\}$
v	class values, $v = (-1, -\frac{1}{2}, 0, \frac{1}{2}, 1)$
$s_{d,k}$	signed stance of d on topic k, $s_{d,k} \in [-1, 1]$

Aggregation and tensors.

$w_{d,c}$	attribution weight: 1 (full credit) or $1/\lvert A_d \rvert$ (fractional)
$S_{c,k,y}$	support of country c on topic k in year y, Eq. (2.3)
$S_{\cdot,\cdot,y}$	year-y stance slice, $S_{\cdot,\cdot,y} \in \mathbb{R}^{C \times \lvert K \rvert}$
$O_{c,k,y}$	opposition of country c on topic k in year y

For these tensors, high sparsity is expected because absence of evidence is recorded as missing, not as zero.

Stance classifier (Chapter 1). For topic k, let h_k^+, h_k^0, h_k^- be the for-, neutral-, and against-pole hypotheses. For passage set d and hypothesis h, the debiased evidence is the null-premise-corrected log-ratio,

$$\lambda(d, h) = \text{clip}\Big(\log P(\text{entail} \mid d, h) - \log P(\text{entail} \mid \varnothing, h),\ \pm 5 \Big),$$

this removes each hypothesis's marginal acceptability. The three-pole distribution combines entailment of each pole with contradiction of its opposite,

$$\tilde{\pi}_{d,k} = \text{softmax} \begin{pmatrix} \lambda_{\text{ent}}(h_k^+) + \lambda_{\text{con}}(h_k^-) \\ \lambda_{\text{ent}}(h_k^0) \\ \lambda_{\text{ent}}(h_k^-) + \lambda_{\text{con}}(h_k^+) \end{pmatrix},$$

these values are expanded to the five ordinal classes $\pi_{d,k}$ and reduced to the scalar stance by the expected value of Equation 2.2.

Appendix D: Acknowledgement of AI Use

I have used Claude Code for boilerplate and scaffolding code in the implementation of the pipelines and scripts. All architectural decisions, methodological choices, and algorithm design were specified by me in the input prompts. I also reviewed and edited generated code. I have also used Consensus AI and the Claude app to fetch the most relevant scientific papers from the available literature, based on my own ideas, keywords, and criteria. Afterwards, I read the papers and drew conclusions myself. Finally, I used Claude Code as a writing assistant for this document, to help with grammar, readability, clarity and LaTeX document formatting. All research questions, project design, ideas, interpretation of results, and selection of literature are my own.